Inversion Sorting

Time limit: 1.00 s **Memory limit:** 512 MB

There is a hidden permutation $a_1, a_2, \ldots, a_n$ of integers $1, 2, \ldots, n$. Your task is to sort the permutation by reversing subarrays.

On each turn, you can reverse a subarray of the permutation. After that, you will be reported the number of inversions in the permutation. If the number of inversions is 0 (i.e., the permutation is sorted), you win.

Interaction

This is an interactive problem. Your code will interact with the grader using standard input and output. You should start by reading a single integer n: the length of the permutation.

On your turn, print two integers i and j: reverse the subarray between indices i and j.

After this, the next input line has a single integer: the number of inversions after the operation. If the number is 0, you win and your program must terminate after this.

Constraints

- $1 \leq n \leq 1000$
- you can make at most $4n$ operations

Example

```
3
1 2
1
2 3
0
```

Explanation: Here the initial permutation is $[3, 1, 2]$. After the first operation the permutation is $[1, 3, 2]$ and the number of inversions is 1. After the second operation the permutation is $[1, 2, 3]$ and the number of inversions is 0.